UNIVERSITY OF CAMBRIDGE

NST1
NATURAL SCIENCES TRIPOS Part IB

Thursday 29 May 2018 1:30 pm to 2:30 pm

Paper PHA/1

PHARMACOLOGY (1)

*Answer **all ten** questions. Note that you have only six minutes for each question, so apportion your time accordingly*

*Start **each** question on a new page*

*Write your **candidate number** on the booklet*

Calculator – students are permitted to bring an approved calculator

STATIONARY REQUIREMENTS

1 x 20-page examination booklet

You may not start to read the questions printed on the subsequent pages of this question paper until instructed that you may do so by the invigilator

1. With the aid of a diagram, describe the G protein cycle.

2. Describe the mechanisms of action of three different classes of drugs used to treat type II diabetes mellitus.

3. How do anticholinesterases act?

4. Briefly discuss mechanisms of action of antibiotics acting on bacterial protein biosynthesis.

5. Draw a diagram to highlight how drugs target the hallmarks of cancer.

6. Explain with graphs the meaning of first-order and zero-order kinetics of drug elimination and the implications of zero-order kinetics when considering a dosage regimen.

7. Describe five drugs acting by distinct mechanisms of action that are used to treat epilepsy.

8. How do diuretics increase sodium ion excretion?

9. Summarise the mechanisms of action of anti-platelet drugs.

10. How do opioids produce analgesia?

END OF PAPER

UNIVERSITY OF CAMBRIDGE

NST1
NATURAL SCIENCES TRIPOS Part IB

Friday 01 June 2018 9:00 to 12:00

Paper PHA/2

PHARMACOLOGY (2)

*Answer **five** questions.*

*Start **each** question in a new booklet*

*Write your **candidate number** on each booklet*

*List the numbers of **all** questions answered on **each** booklet, then circle the number of the question answered in that booklet*

Calculator – students are permitted to bring an approved calculator

STATIONARY REQUIREMENTS

5 x 8-page examination booklet

You may not start to read the questions printed on the subsequent pages of this question paper until instructed that you may do so by the invigilator

1. Discuss the desensitisation of G protein-coupled receptors (GPCRs).
2. Discuss the function of non-adrenergic non-cholinergic (NANC) neurotransmission in the sympathetic and parasympathetic nervous systems.
3. Discuss mechanisms of action of antiparasitic agents.
4. Compare and contrast classical cytotoxic cancer therapy and emerging targeted cancer therapy approaches.
5. Discuss the barriers faced by drugs upon oral administration.
6. Discuss the mechanisms of action of drugs used to treat depression.
7. Compare and contrast the pharmacological properties of different beta adrenoceptor blockers. How do these properties affect their clinical use?
8. Describe how *in silico* approaches can help in discovering and developing novel small molecules against valid therapeutic targets.
9. Discuss the mechanisms of action of drugs used to treat asthma and chronic obstructive pulmonary disorder (COPD).

END OF PAPER